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Techniques for machine learning model selection for domain generalization

Abstract

A computing device may perform training of a set of machine learning models on a first data set associated with a first domain. In some examples, the training may include, for each machine learning model of the set of machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of a set of iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations. The computing device may select a set of model states that are generated during the training of the plurality of machine learning models based on a validation performance of the set of model states performed during the training. The computing device may then generate an ensembled machine learning model by aggregating the set of machine learning models corresponding to the set of selected model states.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
12009071	12/2023	Kim	N/A	G16H 50/70
2022/0121884	12/2021	Zadeh	N/A	G06N 3/006
2022/0253734	12/2021	Aghdasi	N/A	G01N 33/383
2023/0305904	12/2022	Lowe	N/A	G06F 11/3414

OTHER PUBLICATIONS

NPL Aprit Ensemble of Averages Inventor Pub Oct. 2021. cited by examiner

NPL Brownlee A Gentle Introduction to Ensemble Learning Algorithms 2021. cited by examiner

NPL Brownlee How to Develop a Weighted Average Ensemble 2020. cited by examiner

NPL Brownlee Moving Average Smoothing for Data Preparation Time Series 2020. cited by examiner

NPL Dong A survey on ensemble learning 2019. cited by examiner

NPL Ganaie Ensemble deep learning_A review Mar. 2022. cited by examiner

NPL Gulrajani In Search of Lost Domain Generalization 2020. cited by examiner

NPL Izmailov Averaging Weights Leads to Wider Optima and Better Generalization 2019. cited by examiner

NPL Jain Parallelizing SGD for Least Squares Regression 2018. cited by examiner

NPL Kandaswamy Deep Transfer Learning Ensemble 2015. cited by examiner

NPL Lim Ensemble Averaging Mar. 2022. cited by examiner

NPL Liu Ensemble Transfer Learning Algorithm 2017. cited by examiner

NPL Muecke Beating SGD Saturation with Tail Averaging and Minibatching 2019. cited by examiner

NPL Prado Forecasting based on ensemble Autoregressive Moving Average 2020. cited by examiner

NPL Tao Deep Neural Network Ensembles 2019. cited by examiner

NPL Zhao Transfer Learning with Ensemble 2018. cited by examiner

NPL Zhu Ensemble transfer learning 2020. cited by examiner

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Background/Summary

FIELD OF TECHNOLOGY

(1) The present disclosure relates generally to database systems and data processing, and more

specifically to techniques for machine learning model selection for domain generalization.

BACKGROUND

(2) A cloud platform (i.e., a computing platform for cloud computing) may be employed by many users to store, manage, and process data using a shared network of remote servers. Users may develop applications on the cloud platform to handle the storage, management, and processing of data. In some cases, the cloud platform may utilize a multi-tenant database system. Users may access the cloud platform using various user devices (e.g., desktop computers, laptops, smartphones, tablets, or other computing systems, etc.).

(3) In one example, the cloud platform may support customer relationship management (CRM) solutions. This may include support for sales, service, marketing, community, analytics, applications, and the Internet of Things. A user may utilize the cloud platform to help manage contacts of the user. For example, managing contacts of the user may include analyzing data, storing and preparing communications, and tracking opportunities and sales.

(4) Mathematical models (such as machine learning models) may include model parameters. A model parameter may be a configuration variable that is internal to the mathematical model. Such model parameters may be learnt through training of the mathematical models. Machine learning models may work well on data from the same domain as the training data. However, some techniques may lack the ability to train a mathematical model to account for domain generalization. For example, domain generalization may refer to when data during inference comes from a distribution that is different from the training distribution.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates an example of data processing at an application server system that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure.

(2) FIG. 2 illustrates an example of a system that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure.

(3) FIG. 3 illustrates an example of a process flow that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure.

(4) FIG. 4 shows a block diagram of an apparatus that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure.

(5) FIG. 5 shows a block diagram of a machine learning component that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure.

(6) FIG. 6 shows a diagram of a system including a device that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure.

(7) FIGS. 7 through 10 show flowcharts illustrating methods that support techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

(8) Some systems (e.g., artificial intelligence systems supporting customer relationship management (CRM) and one or more datasets) may support a user-friendly, interactive data analytics application. Such an application may receive a request to run one or more mathematical models (or machine learning models) on different data sets. As one example, a user may input a request to run or develop a mathematical model into a data analytics application running on a user device. In some cases, the data analytics application on the user device may transmit the request to

a server (such as an application server). Developing a mathematical model (or machine learning model) may include training the mathematical model. Machine learning models may work well on data having the same domain as the training data. Domain generalization aims at learning predictors that generalize well on data sampled from test distributions that are different from the training distribution.

(9) In domain generalization settings, machine learning models trained on a given set of training domains may have unreliable performance on distribution shifted test domains, and may have stochasticity in performance. In particular, machine learning models trained on a given set of training domains may not be able to perform on distribution shifted test domains. This makes deploying machine learning models (e.g., deep learning models) unreliable in real world settings (settings where the training data and the data used for inference belong to different domains). The techniques depicted herein provide for a protocol for averaging model parameters along an optimization path, starting early during training.

(10) Aspects of the present disclosure describe a training mechanism that takes an average of machine learning model parameters along an optimization path such that the resulting trained machine learning models benefit from the mix of learned knowledge and may improve their performance when being extended to different domains. In particular, when training on a set of data points, the techniques provided herein describe maintaining a moving average of a vector of parameters on a per model basis. That is, a system may train each machine learning model using a vector of parameters for each iteration. For a subsequent iteration, a moving average of the vector of parameters of the model being trained may be maintained during the training process. Instead of calculating the moving average starting from initialization of the model, the techniques provide for starting the averaging after a number of iterations to during training (tail averaging), and maintaining the moving average until the end of training. For each machine learning model, the system may select a moving average state of the vector of parameters having the highest performance (to reduce overfitting). Next, the system may perform ensembling of the moving average models (averaging the states for each machine learning model) to generate a an ensembled machine learning model. In particular, the system generates the ensembled machine learning model by aggregating a set of machine learning models corresponding to a set of selected model states. The system then performs an inference based on deploying the final model on a data set having a different domain than the training data. For instance, the system may perform a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain. In some cases, an output of the ensembled machine learning model may be a dimension-wise average of respective outputs from the set of machine learning models in the ensembled machine learning model.

(11) Thus, the techniques depicted herein provide for significantly boosting domain generalization and diminishing the impact of stochasticity by improving the rank correlation between the in-domain validation accuracy and out-domain test accuracy, which may be utilized for reliable model selection.

(12) Aspects of the disclosure are initially described in the context of an environment supporting an on-demand database service. Aspects of the disclosure are further described in the context of a system and a process flow. Aspects of the disclosure are further illustrated by and described with reference to apparatus diagrams, system diagrams, and flowcharts that relate to techniques for machine learning model selection for domain generalization.

(13) FIG. 1 illustrates an example of a system **100** for cloud computing that supports techniques for machine learning model selection for domain generalization in accordance with various aspects of the present disclosure. The system **100** includes cloud clients **105**, contacts **110**, cloud platform **115**, and data center **120**. Cloud platform **115** may be an example of a public or private cloud network. A cloud client **105** may access cloud platform **115** over network connection **135**. The network may implement transfer control protocol and internet protocol (TCP/IP), such as the

Internet, or may implement other network protocols. A cloud client **105** may be an example of a user device, such as a server (e.g., cloud client **105-a**), a smartphone (e.g., cloud client **105-b**), or a laptop (e.g., cloud client **105-c**). In other examples, a cloud client **105** may be a desktop computer, a tablet, a sensor, or another computing device or system capable of generating, analyzing, transmitting, or receiving communications. In some examples, a cloud client **105** may be operated by a user that is part of a business, an enterprise, a non-profit, a startup, or any other organization type.

(14) A cloud client **105** may interact with multiple contacts **110**. The interactions **130** may include communications, opportunities, purchases, sales, or any other interaction between a cloud client **105** and a contact **110**. Data may be associated with the interactions **130**. A cloud client **105** may access cloud platform **115** to store, manage, and process the data associated with the interactions **130**. In some cases, the cloud client **105** may have an associated security or permission level. A cloud client **105** may have access to some applications, data, and database information within cloud platform **115** based on the associated security or permission level, and may not have access to others.

(15) Contacts **110** may interact with the cloud client **105** in person or via phone, email, web, text messages, mail, or any other appropriate form of interaction (e.g., interactions **130-a**, **130-b**, **130-c**, and **130-d**). The interaction **130** may be a business-to-business (B2B) interaction or a business-to-consumer (B2C) interaction. A contact **110** may also be referred to as a customer, a potential customer, a lead, a client, or some other suitable terminology. In some cases, the contact **110** may be an example of a user device, such as a server (e.g., contact **110-a**), a laptop (e.g., contact **110-b**), a smartphone (e.g., contact **110-c**), or a sensor (e.g., contact **110-d**). In other cases, the contact **110** may be another computing system. In some cases, the contact **110** may be operated by a user or group of users. The user or group of users may be associated with a business, a manufacturer, or any other appropriate organization.

(16) Cloud platform **115** may offer an on-demand database service to the cloud client **105**. In some cases, cloud platform **115** may be an example of a multi-tenant database system. In this case, cloud platform **115** may serve multiple cloud clients **105** with a single instance of software. However, other types of systems may be implemented, including—but not limited to—client-server systems, mobile device systems, and mobile network systems. In some cases, cloud platform **115** may support CRM solutions. This may include support for sales, service, marketing, community, analytics, applications, and the Internet of Things. Cloud platform **115** may receive data associated with contact interactions **130** from the cloud client **105** over network connection **135**, and may store and analyze the data. In some cases, cloud platform **115** may receive data directly from an interaction **130** between a contact **110** and the cloud client **105**. In some cases, the cloud client **105** may develop applications to run on cloud platform **115**. Cloud platform **115** may be implemented using remote servers. In some cases, the remote servers may be located at one or more data centers **120**.

(17) Data center **120** may include multiple servers. The multiple servers may be used for data storage, management, and processing. Data center **120** may receive data from cloud platform **115** via connection **140**, or directly from the cloud client **105** or an interaction **130** between a contact **110** and the cloud client **105**. Data center **120** may utilize multiple redundancies for security purposes. In some cases, the data stored at data center **120** may be backed up by copies of the data at a different data center (not pictured).

(18) Subsystem **125** may include cloud clients **105**, cloud platform **115**, and data center **120**. In some cases, data processing may occur at any of the components of subsystem **125**, or at a combination of these components. In some cases, servers may perform the data processing. The servers may be a cloud client **105** or located at data center **120**.

(19) Domain generalization in training of machine learning models may be used to learn predictors that generalize well on data sampled from test distributions that are different from the training

distribution. In some examples, deep learning models may demonstrate poor performance at this form of generalization, and excel primarily in a setting including independent and identically distributed random variables.

(20) Some machine learning models may be designed to mitigate this drawback using empirical risk minimization along with a model selection techniques (i.e., early stopping using validation set), using a subset of data from all the training domains. Such machine learning models may largely match or even outperform the performance of existing domain generalization algorithms. However, such machine learning models may not have a high reliability associated with model selection. In particular, the out-domain performance may vary greatly along the optimization trajectory of a machine learning model during training, even though the in-domain performance may not vary. This instability hurts the reliability of model selection, and may become a problem in realistic settings where test domain data is unavailable. In particular, without being able to generalize performance across multiple domains, deploying a machine learning model may cause a rank correlation between in-domain validation accuracy and out-domain test accuracy to be weak.

(21) In contrast, the system **100** provides for model averaging that both boosts domain generalization, and mitigates performance instability of deep models on out-domain data, specifically with respect to in-domain validation data. This makes model selection more reliable. In addition, the system **100** supports ensembling of moving average models to mitigate the lack of performance in domain generalizations settings. Additionally or alternatively, the system **100** may support generation of an ensembled machine learning model that further boosts performance of machine learning models of test data having a domain different from training data.

(22) The system **100** may first perform training of a set of machine learning models on a first data set associated with a first domain. The set of machine learning models may include respective sets of parameters that are updated across a set of iterations during the training. According to one or more aspects of the present disclosure, as part of the training, for each machine learning model of the set of machine learning models, the subsystem **125** may input, as values for a set of parameters of the respective sets of parameters and for an iteration of the set of iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations. The subsystem **125** may then select a set of model states that are generated during the training of the set of machine learning models. In some cases, the set of model states may be selected based on a validation performance of the set of model states performed during the training. That is, the subsystem **125** may determine a model state corresponding to each machine learning model, that provides a high performance.

(23) Such moving average models may have a more stable out of domain performance compared to unaveraged models. The rank correlation between in-domain validation accuracy and out-domain test accuracy along the optimization trajectory may also be better for moving average models. Thus, model selection may be more reliable when the subsystem **125** makes predictions on a validation set using the moving average models. In particular, starting averaging early during training, may boost domain generalization. However, since the rank correlation may be poor between validation and test accuracy of independently trained models, it may be difficult to discover a machine learning model having high performance (for out-domain performance) from a pool of independently trained models, based on their in-domain validation performance. To enhance the performance of the machine learning model on out of domain data, the subsystem **125** may generate an ensemble of moving average models. In particular, the subsystem **125** may generate an ensembled machine learning model by aggregating the set of machine learning models corresponding to the set of selected model states (the optimal states for each machine learning model). The frequency of model averaging may not have a significant impact on performance, unless sampling is done at too large intervals.

(24) The ensemble of moving average models may outperform an ensemble of unaveraged models. The subsystem **125** may perform a machine learning prediction using the ensembled machine

learning model on a second data set associated with a second domain different from the first domain. In such case, an output of the ensembled machine learning model may be a dimension-wise average of respective outputs from the set of machine learning models in the ensembled machine learning model. An ablation analysis may indicate that the rank correlation between in-domain validation performance and out-domain test performance is also better for the ensemble of average models.

(25) In some examples, the subsystem **125** may train the set of machine learning models using image data. Such image data may include a set of images each having a set of parameters (e.g., brightness, contrast, sharpness, color saturation, etc.). The subsystem **125** may generate an ensembled machine learning model and then use the ensembled machine learning model to predict images having a different set of parameters. For instance, the ensembled machine learning model may be trained using photographs of cats and the ensembled machine learning model may be used to predict cartoons of cats. In such cases, the ensembled machine learning model may improve the baseline performance (e.g., around 4%-5% test accuracy gain on average).

(26) It should be appreciated by a person skilled in the art that one or more aspects of the disclosure may be implemented in a system **100** to additionally or alternatively solve other problems than those described herein. Furthermore, aspects of the disclosure may provide technical improvements to “conventional” systems or processes as described herein. However, the description and appended drawings only include example technical improvements resulting from implementing aspects of the disclosure, and accordingly do not represent all of the technical improvements provided within the scope of the claims.

(27) FIG. 2 illustrates an example of a system **200** that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure. The system **200** may be an example of a system **100**, as described with reference to FIG. 1. The computing device **205** may support a CRM platform, in some examples. For example, the computing device **205** may be an example of an application server, a database server, a cloud-based server or service, a worker server, a server cluster, a virtual machine, a container, or any combination of these or other computing devices. The user device **210** may be an example of a contact **110**, such as a laptop, a desktop computer, a tablet, a smart phone, a smart watch, a smart television, or any other user-operated device. The user device **210** may access an application supporting to access the computing device **205**. For example, a user operating the user device **210** may log into the application, and the computing device **205** may send, for display in the user interface of the user device **210**, information relating to a platform for generating or accessing an ensembled machine learning model. In some alternative examples, the user device **210** may host the application to generate the ensembled machine learning model locally at the user device **210**, and one or more operations described herein with respect to the computing device **205** may alternatively be performed by the user device **210**. The system **200** may use techniques described herein to improve out-domain performance by machine learning model selection for domain generalization.

(28) The system **200** may perform model averaging followed by ensembling machine learning models. For a given supervised learning objective function, $f_{\text{sub}}(\theta, \text{Math.})$ may denote a deep network being optimized using gradient based optimizer, where θ denotes the parameters of this model. $f_{\text{sub}}(\theta)$ may be referred to as an online model, or unaveraged model. The output of $f_{\text{sub}}(\theta, \text{Math.})$ may be a vector of K logits corresponding to the K classes in the supervised task. The computing device **205** may use an input of training data **220** to train the machine learning model **225**. The computing device **205** may perform training of a set of machine learning models on a first data set associated with a first domain (e.g., training data **220**). The set of machine learning models may include respective sets of parameters that are updated across a set of iterations during the training. As part of the training, for each machine learning model, the computing device **205** may input, as values for a set of parameters of the respective sets of parameters and for an

iteration of the set of iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations. That is, the computing device **205** may maintain a moving average of the machine learning model's **225** parameters. The deep network (e.g., machine learning model **225**) whose parameters are set to be this moving average may be referred to as the moving average model, or more specifically simple moving average model. The parameters of this model may be denoted by $\{\text{circumflex over } (\theta)\}$.

(29) As the simple moving average of the online model, instead of calculating the moving average starting from initialization, the computing device **205** may instead start after a number of iterations t_0 during training (tail averaging), and maintain the moving average until the end of training. In some examples, the computing device **205** may choose t_0 to be close, but not equal to the initialization. For instance, the computing device **205** may start a calculation of the moving average of the set of parameters after a configured number of iterations from a starting iteration of the set of iterations.

(30) At any iteration t , the computing device **205** denote the following:

$$(31) \quad \hat{\theta}_t = \left\{ \frac{t}{t - t_0 + 1} \cdot \hat{\theta}_{t+1} + \frac{1}{t - t_0 + 1} \cdot \theta_t \right\}$$

(32) if $t \leq t_0$, where $\theta_{\text{sub}.t}$ is the online model's state at iteration t . Further, at iteration t , the computing device **205** may calculate validation performance using $\{\text{circumflex over } (\theta)\}_{\text{sub}.t}$, and not $\theta_{\text{sub}.t}$. The benefit of doing so is that the rank correlation between in-domain validation accuracy and out-domain test accuracy is better when predictions are made using $\{\text{circumflex over } (\theta)\}_{\text{sub}.t}$. This makes model selection more reliable for domain generalization. Finally, for a given run, model selection may select $\{\text{circumflex over } (\theta)\}_{\text{sub}.t}$ for making test set predictions, such that $\{\text{circumflex over } (\theta)\}_{\text{sub}.t}$ achieves high validation performance.

(33) According to one or more aspects, the computing device **205** may select a set of model states **240** that are generated during the training of the set of machine learning models (e.g., machine learning model **225**). In some examples, the computing device **205** may select the set of model states **240** based on a validation performance of the set of model states **240** performed during the training. The computing device **205** may determine, for one or more iterations of the set of iterations, a validation performance value associated with a current state of a machine learning model. The computing device **205** may select a model state for one or more machine learning models based on a highest validation performance value for the corresponding machine learning model across all iterations of a set of iterations.

(34) In some examples, the computing device **205** may select a set of model states that are generated during the training of the set of machine learning models, where the set of model states are selected based on a validation performance of the set of model states performed during the training. The computing device **205** may input the training data **220** to the machine learning model **225** throughout multiple iterations. The computing device **205** may calculate a validation performance value for each iteration. As depicted in the example of FIG. 2, the computing device **205** may train a first machine learning model **245**, a second machine learning model **250**, and a third machine learning model **255**. The computing device **205** may determine that the state S_3 for the first machine learning model **245**, the state S_2 for the second machine learning model **250**, and the state S_3 for the third machine learning model **255** correspond to the highest validation performance scores for each machine learning model. The computing device **205** may thus use the state S_3 for the first machine learning model **245**, the state S_2 for the second machine learning model **250**, and the state S_3 for the third machine learning model **255** for generating an ensembled machine learning model **230**.

(35) As depicted herein, the computing device **205** generates an ensembled machine learning model to handle domain generalization settings. In some examples, a framework for evaluation in the domain generalization setting which accounts for randomness due to seed and hyperparameter

values, and recommend reporting the average test accuracy over all the runs computed using a model selection criteria. It may be desirable to have a single predictor that has a high accuracy. Ensembles combine predictions from multiple models by exploiting function diversity. However, simple ensembles may suffer from instability in the domain generalization setting. Building on the observations discussed herein, the computing device **205** may generate an ensemble of moving average models. In some examples, the computing device **205** may determine an experiment with ensemble of multiple independently trained models (i.e., with different hyper-parameters and seeds). When each of these models are moving average models from their corresponding runs, this ensemble may be referred to as the ensemble of averages. The computing device **205** may predict the class **9** predicted by an ensemble of averages for an input x is according to the formula:

$$(36) \hat{y} = \operatorname{argmax}_k \operatorname{Softmax}(\frac{1}{E} \sum_{i=1}^E \operatorname{Math}.\hat{f}_i^{(k)}(x))$$

(37) where E is the total number of models in the ensemble, $\{\hat{\theta}\}_{i=1}^E$ denotes the parameters of the i th moving average model, and the super-script $\operatorname{Math}.\hat{f}_i^{(k)}$ denotes the k th element of the vector argument. Finally, the state $\{\hat{\theta}\}_{i=1}^E$ of the i th moving average model used in the ensemble may be selected from its corresponding run using its in-domain validation set performance.

(38) The computing device **205** may then perform a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain. An output of the ensembled machine learning model may be or otherwise include a dimension-wise average of respective outputs from the set of machine learning models in the ensembled machine learning model. Thus, the computing device **205** may input the test data to the set of machine learning models are the selected model states (e.g., the state $S3$ for the first machine learning model **245**, the state $S2$ for the second machine learning model **250**, and the state $S3$ for the third machine learning model **255**). The computing device **205** may then perform an average of the outputs from each model to generate a final prediction. The computing device **205** may indicate the final prediction in results **235** to the user device **210**.

(39) Techniques depicted herein may be implemented to illustrate that instead of ensembling unaveraged models, ensembling moving average models from different runs increases stability and further boosts performance. In one examples, when using a ResNet-50 pre-trained on a first dataset having 4 domains, 7 classes, and 9,991 images, this ensemble of averages may achieve 88.6% accuracy on the first dataset, 79.1% on the third dataset having 4 domains, 5 classes, and 10,729 images, 72.5% on the fourth dataset having 4 domains, 65 classes, and 15,588 images, 52.3% on the second dataset, and 47.4% on the fifth dataset having 4 domains, 10 classes, and 24,788 images, an average of 68.0% (which is more than empirical risk minimization (ERM) algorithm without model averaging by 4%). For a model that is pre-trained on a larger dataset, the ensemble of averages achieve an average accuracy of 72.7%, which is greater than its corresponding ERM baseline by 5%.

(40) Comparing ERM over different datasets, they may perform similarly with a smaller hyper-parameter space. A comparison between stochastic weighted averaging densely (SWAD) and ERM with model average may indicate similar performance. However, ensemble of averages protocol retains the advantage of not tuning any hyper-parameters while SWAD has 3 additional ones to be tuned separately in addition to the optimization hyper-parameters. Finally, the ensemble of averages may outperform all existing results: ERM by 4% and SWAD by 1:1%, while non-ensemble models may report the average test accuracy of multiple models, the test accuracy for ensemble of averages may be achieved by a single predictor that combines the output of multiple models.

(41) FIG. 3 illustrates an example of a process flow **300** that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure. The process flow **300** may be implemented by a system including one or more

computing devices **305** and one or more user devices **310**. The computing device **305** may host a platform for generating an ensembles machine learning model. The user device **310** may include user interfaces that can display information corresponding to the results of the ensembled machine learning model, as described herein with reference to FIGS. **1** through **2**. Alternative examples of the following may be implemented, where some processes are performed in a different order than described or are not performed at all. In some examples, processes may include additional features not mentioned below, or further processes may be added. Additionally or alternatively, operations described herein with reference to the computing device **305** may instead be performed by the host user device **310**.

(42) In some examples, at **315**, the computing device **305** may perform training of a set of machine learning models on a first data set associated with a first domain. In some examples, the set of machine learning models may include respective sets of parameters that are updated across a plurality of iterations during the training. The training may include, for each machine learning model of the set of machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the set of iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations.

(43) The computing device **305** may start a calculation of the moving average of the set of parameters after a configured number of iterations from a starting iteration of the set of iterations. In some cases, the set of machine learning models may be trained using a gradient based technique.

(44) At **320**, the computing device **305** may select a set of model states that a generated during the training of the set of machine learning models. In some examples, the set of model states may be selected based on a validation performance of the set of model states performed during the training. In some examples, the computing device **305** may determine, for a first iteration, a first validation performance value associated with a first state of a machine learning model of the set of machine learning models. The computing device **305** may determine, for a second iteration following the first iteration, a second validation performance value associated with a second state of the machine learning model of the set of machine learning models. In some examples, the computing device **305** may select the first state of the machine learning model for generation of the ensembled machine learning model based on determining that the second validation performance value is less than the first validation performance value.

(45) At **325**, the computing device **305** may generate an ensembled machine learning model by aggregating the plurality of machine learning models corresponding to the set of selected model states. At **330**, the computing device **305** may perform a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain. In some examples, an output of the ensembled machine learning model may be a dimension-wise average of respective outputs from the set of machine learning models in the ensembled machine learning model. The respective outputs from the set of machine learning models may include a respective vector for each machine learning model and the output of the ensembled machine learning model may include the dimension-wise average of the respective vector.

(46) FIG. **4** shows a block diagram **400** of a device **405** that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure. The device **405** may include an input component **410**, an output component **415**, and a machine learning component **420**. The device **405** may also include a processor. Each of these components may be in communication with one another (e.g., via one or more buses).

(47) The input component **410** may manage input signals for the apparatus **405**. For example, the input component **410** may identify input signals based on an interaction with a modem, a keyboard, a mouse, a touchscreen, or a similar device. These input signals may be associated with user input or processing at other components or devices. In some cases, the input component **410** may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®,

LINUX®, or another known operating system to handle input signals. The input component **410** may send aspects of these input signals to other components of the apparatus **405** for processing. For example, the input component **410** may transmit input signals to the machine learning component **420** to support techniques for machine learning model selection for domain generalization. In some cases, the input component **410** may be a component of an I/O controller **610** as described with reference to FIG. 6.

(48) The output component **415** may manage output signals for the apparatus **405**. For example, the output component **415** may receive signals from other components of the apparatus **405**, such as the machine learning component **420**, and may transmit these signals to other components or devices. In some specific examples, the output component **415** may transmit output signals for display in a user interface, for storage in a database or data store, for further processing at a server or server cluster, or for any other processes at any number of devices or systems. In some cases, the output component **415** may be a component of an I/O controller **610** as described with reference to FIG. 6.

(49) For example, the machine learning component **420** may include a training component **425**, a model selection component **430**, a model aggregation component **435**, a prediction component **440**, or any combination thereof. In some examples, the machine learning component **420**, or various components thereof, may be configured to perform various operations (e.g., receiving, monitoring, transmitting) using or otherwise in cooperation with the input component **410**, the output component **415**, or both. For example, the machine learning component **420** may receive information from the input component **410**, send information to the output component **415**, or be integrated in combination with the input component **410**, the output component **415**, or both to receive information, transmit information, or perform various other operations as described herein.

(50) The machine learning component **420** may support machine learning model training in accordance with examples as disclosed herein. The training component **425** may be configured as or otherwise support a means for performing training of a set of multiple machine learning models on a first data set associated with a first domain, where the set of multiple machine learning models includes respective sets of parameters that are updated across a set of multiple iterations during the training, where the training includes, for each machine learning model of the set of multiple machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the set of multiple iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations. The model selection component **430** may be configured as or otherwise support a means for selecting a set of multiple model states that are generated during the training of the set of multiple machine learning models, where the set of multiple model states are selected based on a validation performance of the set of multiple model states performed during the training. The model aggregation component **435** may be configured as or otherwise support a means for generating an ensembled machine learning model by aggregating the set of multiple machine learning models corresponding to the set of multiple selected model states. The prediction component **440** may be configured as or otherwise support a means for performing a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain, where an output of the ensembled machine learning model is a dimension-wise average of respective outputs from the set of multiple machine learning models in the ensembled machine learning model.

(51) FIG. 5 shows a block diagram **500** of a machine learning component **520** that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure. The machine learning component **520** may be an example of aspects of a machine learning component or a machine learning component **420**, or both, as described herein. The machine learning component **520**, or various components thereof, may be an example of means for performing various aspects of techniques for machine learning model

selection for domain generalization as described herein. For example, the machine learning component **520** may include a training component **525**, a model selection component **530**, a model aggregation component **535**, a prediction component **540**, a validation component **545**, a value generation component **550**, or any combination thereof. Each of these components may communicate, directly or indirectly, with one another (e.g., via one or more buses).

(52) The machine learning component **520** may support machine learning model training in accordance with examples as disclosed herein. The training component **525** may be configured as or otherwise support a means for performing training of a set of multiple machine learning models on a first data set associated with a first domain, where the set of multiple machine learning models includes respective sets of parameters that are updated across a set of multiple iterations during the training, where the training includes, for each machine learning model of the set of multiple machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the set of multiple iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations. The model selection component **530** may be configured as or otherwise support a means for selecting a set of multiple model states that are generated during the training of the set of multiple machine learning models, where the set of multiple model states are selected based on a validation performance of the set of multiple model states performed during the training. The model aggregation component **535** may be configured as or otherwise support a means for generating an ensembled machine learning model by aggregating the set of multiple machine learning models corresponding to the set of multiple selected model states. The prediction component **540** may be configured as or otherwise support a means for performing a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain, where an output of the ensembled machine learning model is a dimension-wise average of respective outputs from the set of multiple machine learning models in the ensembled machine learning model.

(53) In some examples, the validation component **545** may be configured as or otherwise support a means for determining, for one or more iterations of the set of multiple iterations, a validation performance value associated with a current state of a machine learning model of the set of multiple machine learning models. In some examples, the model selection component **530** may be configured as or otherwise support a means for selecting a model state for one or more machine learning models of the set of multiple machine learning models based on a highest validation performance value for the corresponding machine learning model across all iterations of the set of multiple iterations.

(54) In some examples, the value generation component **550** may be configured as or otherwise support a means for determining, for a first iteration following the threshold number of previous iterations, a first set of values for the set of parameters of a machine learning model of the set of multiple machine learning models. In some examples, the value generation component **550** may be configured as or otherwise support a means for determining, for a second iteration following the first iteration, a second set of values for the set of parameters of the machine learning model based on the moving average of the set of parameters calculated during the first iteration.

(55) In some examples, the training component **525** may be configured as or otherwise support a means for starting a calculation of the moving average of the set of parameters after a configured number of iterations from a starting iteration of the set of multiple iterations.

(56) In some examples, the validation component **545** may be configured as or otherwise support a means for determining, for a first iteration, a first validation performance value associated with a first state of a machine learning model of the set of multiple machine learning models. In some examples, the validation component **545** may be configured as or otherwise support a means for determining, for a second iteration following the first iteration, a second validation performance value associated with a second state of the machine learning model of the set of multiple machine

learning models. In some examples, the model selection component **530** may be configured as or otherwise support a means for selecting the first state of the machine learning model for generation of the ensembled machine learning model based on determining that the second validation performance value is less than the first validation performance value.

(57) In some examples, the set of multiple machine learning models are trained using a gradient based technique. In some examples, the respective outputs from the set of multiple machine learning models includes a respective vector for each machine learning model and the output of the ensembled machine learning model includes the dimension-wise average of the respective vector.

(58) FIG. **6** shows a diagram of a system **600** including a device **605** that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure. The device **605** may be an example of or include the components of a device **405** as described herein. The device **605** may include components for bi-directional data communications including components for transmitting and receiving communications, such as a machine learning component **620**, an I/O controller **610**, a database controller **615**, a memory **625**, a processor **630**, and a database **635**. These components may be in electronic communication or otherwise coupled (e.g., operatively, communicatively, functionally, electronically, electrically) via one or more buses (e.g., a bus **640**).

(59) The I/O controller **610** may manage input signals **645** and output signals **650** for the device **605**. The I/O controller **610** may also manage peripherals not integrated into the device **605**. In some cases, the I/O controller **610** may represent a physical connection or port to an external peripheral. In some cases, the I/O controller **610** may utilize an operating system such as iOS®, ANDROID®, MS-DOS®, MS-WINDOWS®, OS/2®, UNIX®, LINUX®, or another known operating system. Additionally or alternatively, the I/O controller **610** may represent or interact with a modem, a keyboard, a mouse, a touchscreen, or a similar device. In some cases, the I/O controller **610** may be implemented as part of a processor. In some examples, a user may interact with the device **605** via the I/O controller **610** or via hardware components controlled by the I/O controller **610**.

(60) The database controller **615** may manage data storage and processing in a database **635**. The database **635** may be external to the device **605**, temporarily or permanently connected to the device **605**, or a data storage component of the device **605**. In some cases, a user may interact with the database controller **615**. In some other cases, the database controller **615** may operate automatically without user interaction. The database **635** may be an example of a persistent data store, a single database, a distributed database, multiple distributed databases, a database management system, or an emergency backup database.

(61) Memory **625** may include random-access memory (RAM) and ROM. The memory **625** may store computer-readable, computer-executable software including instructions that, when executed, cause the processor to perform various functions described herein. In some cases, the memory **625** may contain, among other things, a BIOS which may control basic hardware or software operation such as the interaction with peripheral components or devices.

(62) The processor **630** may include an intelligent hardware device (e.g., a general-purpose processor, a DSP, a CPU, a microcontroller, an ASIC, an FPGA, a programmable logic device, a discrete gate or transistor logic component, a discrete hardware component, or any combination thereof). In some cases, the processor **630** may be configured to operate a memory array using a memory controller. In some other cases, a memory controller may be integrated into the processor **630**. The processor **630** may be configured to execute computer-readable instructions stored in memory **625** to perform various functions (e.g., functions or tasks supporting techniques for machine learning model selection for domain generalization).

(63) The machine learning component **620** may support machine learning model training in accordance with examples as disclosed herein. For example, the machine learning component **620** may be configured as or otherwise support a means for performing training of a set of multiple

machine learning models on a first data set associated with a first domain, where the set of multiple machine learning models includes respective sets of parameters that are updated across a set of multiple iterations during the training, where the training includes, for each machine learning model of the set of multiple machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the set of multiple iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations. The machine learning component **620** may be configured as or otherwise support a means for selecting a set of multiple model states that are generated during the training of the set of multiple machine learning models, where the set of multiple model states are selected based on a validation performance of the set of multiple model states performed during the training. The machine learning component **620** may be configured as or otherwise support a means for generating an ensembled machine learning model by aggregating the set of multiple machine learning models corresponding to the set of multiple selected model states. The machine learning component **620** may be configured as or otherwise support a means for performing a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain, where an output of the ensembled machine learning model is a dimension-wise average of respective outputs from the set of multiple machine learning models in the ensembled machine learning model.

(64) By including or configuring the machine learning component **620** in accordance with examples as described herein, the device **605** may support techniques for boosting domain generalization and diminishing the impact of stochasticity by improving the rank correlation between the in-domain validation accuracy and out-domain test accuracy, which may be utilized for reliable model selection.

(65) FIG. 7 shows a flowchart illustrating a method **700** that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure. The operations of the method **700** may be implemented by a computing device or its components as described herein. For example, the operations of the method **700** may be performed by a computing device as described with reference to FIGS. 1 through 6. In some examples, a computing device may execute a set of instructions to control the functional elements of the computing device to perform the described functions. Additionally, or alternatively, the computing device may perform aspects of the described functions using special-purpose hardware.

(66) At **705**, the method may include performing training of a set of multiple machine learning models on a first data set associated with a first domain, where the set of multiple machine learning models includes respective sets of parameters that are updated across a set of multiple iterations during the training, where the training includes, for each machine learning model of the set of multiple machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the set of multiple iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations. The operations of **705** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **705** may be performed by a training component **525** as described with reference to FIG. 5.

(67) At **710**, the method may include selecting a set of multiple model states that are generated during the training of the set of multiple machine learning models, where the set of multiple model states are selected based on a validation performance of the set of multiple model states performed during the training. The operations of **710** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **710** may be performed by a model selection component **530** as described with reference to FIG. 5.

(68) At **715**, the method may include generating an ensembled machine learning model by aggregating the set of multiple machine learning models corresponding to the set of multiple selected model states. The operations of **715** may be performed in accordance with examples as

disclosed herein. In some examples, aspects of the operations of **715** may be performed by a model aggregation component **535** as described with reference to FIG. 5.

(69) At **720**, the method may include performing a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain, where an output of the ensembled machine learning model is a dimension-wise average of respective outputs from the set of multiple machine learning models in the ensembled machine learning model. The operations of **720** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **720** may be performed by a prediction component **540** as described with reference to FIG. 5.

(70) FIG. **8** shows a flowchart illustrating a method **800** that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure. The operations of the method **800** may be implemented by a computing device or its components as described herein. For example, the operations of the method **800** may be performed by a computing device as described with reference to FIGS. **1** through **6**. In some examples, a computing device may execute a set of instructions to control the functional elements of the computing device to perform the described functions. Additionally, or alternatively, the computing device may perform aspects of the described functions using special-purpose hardware.

(71) At **805**, the method may include performing training of a set of multiple machine learning models on a first data set associated with a first domain, where the set of multiple machine learning models includes respective sets of parameters that are updated across a set of multiple iterations during the training, where the training includes, for each machine learning model of the set of multiple machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the set of multiple iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations. The operations of **805** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **805** may be performed by a training component **525** as described with reference to FIG. 5.

(72) At **810**, the method may include selecting a set of multiple model states that are generated during the training of the set of multiple machine learning models, where the set of multiple model states are selected based on a validation performance of the set of multiple model states performed during the training. The operations of **810** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **810** may be performed by a model selection component **530** as described with reference to FIG. 5.

(73) At **815**, the method may include determining, for one or more iterations of the set of multiple iterations, a validation performance value associated with a current state of a machine learning model of the set of multiple machine learning models. The operations of **815** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **815** may be performed by a validation component **545** as described with reference to FIG. 5.

(74) At **820**, the method may include selecting a model state for one or more machine learning models of the set of multiple machine learning models based on a highest validation performance value for the corresponding machine learning model across all iterations of the set of multiple iterations. The operations of **820** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **820** may be performed by a model selection component **530** as described with reference to FIG. 5.

(75) At **825**, the method may include generating an ensembled machine learning model by aggregating the set of multiple machine learning models corresponding to the set of multiple selected model states. The operations of **825** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **825** may be performed by a model aggregation component **535** as described with reference to FIG. 5.

(76) At **830**, the method may include performing a machine learning prediction using the

ensembled machine learning model on a second data set associated with a second domain different from the first domain, where an output of the ensembled machine learning model is a dimension-wise average of respective outputs from the set of multiple machine learning models in the ensembled machine learning model. The operations of **830** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **830** may be performed by a prediction component **540** as described with reference to FIG. 5.

(77) FIG. 9 shows a flowchart illustrating a method **900** that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure. The operations of the method **900** may be implemented by a computing device or its components as described herein. For example, the operations of the method **900** may be performed by a computing device as described with reference to FIGS. 1 through 6. In some examples, a computing device may execute a set of instructions to control the functional elements of the computing device to perform the described functions. Additionally, or alternatively, the computing device may perform aspects of the described functions using special-purpose hardware.

(78) At **905**, the method may include performing training of a set of multiple machine learning models on a first data set associated with a first domain, where the set of multiple machine learning models includes respective sets of parameters that are updated across a set of multiple iterations during the training, where the training includes, for each machine learning model of the set of multiple machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the set of multiple iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations. The operations of **905** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **905** may be performed by a training component **525** as described with reference to FIG. 5.

(79) At **910**, the method may include determining, for a first iteration following the threshold number of previous iterations, a first set of values for the set of parameters of a machine learning model of the set of multiple machine learning models. The operations of **910** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **910** may be performed by a value generation component **550** as described with reference to FIG. 5.

(80) At **915**, the method may include determining, for a second iteration following the first iteration, a second set of values for the set of parameters of the machine learning model based on the moving average of the set of parameters calculated during the first iteration. The operations of **915** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **915** may be performed by a value generation component **550** as described with reference to FIG. 5.

(81) At **920**, the method may include selecting a set of multiple model states that are generated during the training of the set of multiple machine learning models, where the set of multiple model states are selected based on a validation performance of the set of multiple model states performed during the training. The operations of **920** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **920** may be performed by a model selection component **530** as described with reference to FIG. 5.

(82) At **925**, the method may include generating an ensembled machine learning model by aggregating the set of multiple machine learning models corresponding to the set of multiple selected model states. The operations of **925** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **925** may be performed by a model aggregation component **535** as described with reference to FIG. 5.

(83) At **930**, the method may include performing a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain, where an output of the ensembled machine learning model is a dimension-wise average of respective outputs from the set of multiple machine learning models in the

ensembled machine learning model. The operations of **930** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **930** may be performed by a prediction component **540** as described with reference to FIG. 5.

(84) FIG. **10** shows a flowchart illustrating a method **1000** that supports techniques for machine learning model selection for domain generalization in accordance with aspects of the present disclosure. The operations of the method **1000** may be implemented by a computing device or its components as described herein. For example, the operations of the method **1000** may be performed by a computing device as described with reference to FIGS. **1** through **6**. In some examples, a computing device may execute a set of instructions to control the functional elements of the computing device to perform the described functions. Additionally, or alternatively, the computing device may perform aspects of the described functions using special-purpose hardware.

(85) At **1005**, the method may include performing training of a set of multiple machine learning models on a first data set associated with a first domain, where the set of multiple machine learning models includes respective sets of parameters that are updated across a set of multiple iterations during the training, where the training includes, for each machine learning model of the set of multiple machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the set of multiple iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations. The operations of **1005** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1005** may be performed by a training component **525** as described with reference to FIG. 5.

(86) At **1010**, the method may include starting a calculation of the moving average of the set of parameters after a configured number of iterations from a starting iteration of the set of multiple iterations. The operations of **1010** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1010** may be performed by a training component **525** as described with reference to FIG. 5.

(87) At **1015**, the method may include selecting a set of multiple model states that are generated during the training of the set of multiple machine learning models, where the set of multiple model states are selected based on a validation performance of the set of multiple model states performed during the training. The operations of **1015** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1015** may be performed by a model selection component **530** as described with reference to FIG. 5.

(88) At **1020**, the method may include generating an ensembled machine learning model by aggregating the set of multiple machine learning models corresponding to the set of multiple selected model states. The operations of **1020** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1020** may be performed by a model aggregation component **535** as described with reference to FIG. 5.

(89) At **1025**, the method may include performing a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain, where an output of the ensembled machine learning model is a dimension-wise average of respective outputs from the set of multiple machine learning models in the ensembled machine learning model. The operations of **1025** may be performed in accordance with examples as disclosed herein. In some examples, aspects of the operations of **1025** may be performed by a prediction component **540** as described with reference to FIG. 5.

(90) A method for machine learning model training is described. The method may include performing training of a set of multiple machine learning models on a first data set associated with a first domain, where the set of multiple machine learning models includes respective sets of parameters that are updated across a set of multiple iterations during the training, where the training includes, for each machine learning model of the set of multiple machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration

of the set of multiple iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations, selecting a set of multiple model states that are generated during the training of the set of multiple machine learning models, where the set of multiple model states are selected based on a validation performance of the set of multiple model states performed during the training, generating an ensembled machine learning model by aggregating the set of multiple machine learning models corresponding to the set of multiple selected model states, and performing a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain, where an output of the ensembled machine learning model is a dimension-wise average of respective outputs from the set of multiple machine learning models in the ensembled machine learning model.

(91) An apparatus for machine learning model training is described. The apparatus may include a processor, memory coupled with the processor, and instructions stored in the memory. The instructions may be executable by the processor to cause the apparatus to perform training of a set of multiple machine learning models on a first data set associated with a first domain, where the set of multiple machine learning models includes respective sets of parameters that are updated across a set of multiple iterations during the training, where the training includes, for each machine learning model of the set of multiple machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the set of multiple iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations, select a set of multiple model states that are generated during the training of the set of multiple machine learning models, where the set of multiple model states are selected based on a validation performance of the set of multiple model states performed during the training, generate an ensembled machine learning model by aggregating the set of multiple machine learning models corresponding to the set of multiple selected model states, and perform a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain, where an output of the ensembled machine learning model is a dimension-wise average of respective outputs from the set of multiple machine learning models in the ensembled machine learning model.

(92) Another apparatus for machine learning model training is described. The apparatus may include means for performing training of a set of multiple machine learning models on a first data set associated with a first domain, where the set of multiple machine learning models includes respective sets of parameters that are updated across a set of multiple iterations during the training, where the training includes, for each machine learning model of the set of multiple machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the set of multiple iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations, means for selecting a set of multiple model states that are generated during the training of the set of multiple machine learning models, where the set of multiple model states are selected based on a validation performance of the set of multiple model states performed during the training, means for generating an ensembled machine learning model by aggregating the set of multiple machine learning models corresponding to the set of multiple selected model states, and means for performing a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain, where an output of the ensembled machine learning model is a dimension-wise average of respective outputs from the set of multiple machine learning models in the ensembled machine learning model.

(93) A non-transitory computer-readable medium storing code for machine learning model training is described. The code may include instructions executable by a processor to perform training of a set of multiple machine learning models on a first data set associated with a first domain, where the set of multiple machine learning models includes respective sets of parameters that are updated across a set of multiple iterations during the training, where the training includes, for each machine

learning model of the set of multiple machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the set of multiple iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations, select a set of multiple model states that are generated during the training of the set of multiple machine learning models, where the set of multiple model states are selected based on a validation performance of the set of multiple model states performed during the training, generate an ensembled machine learning model by aggregating the set of multiple machine learning models corresponding to the set of multiple selected model states, and perform a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain, where an output of the ensembled machine learning model is a dimension-wise average of respective outputs from the set of multiple machine learning models in the ensembled machine learning model.

(94) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining, for one or more iterations of the set of multiple iterations, a validation performance value associated with a current state of a machine learning model of the set of multiple machine learning models and selecting a model state for one or more machine learning models of the set of multiple machine learning models based on a highest validation performance value for the corresponding machine learning model across all iterations of the set of multiple iterations.

(95) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining, for a first iteration following the threshold number of previous iterations, a first set of values for the set of parameters of a machine learning model of the set of multiple machine learning models and determining, for a second iteration following the first iteration, a second set of values for the set of parameters of the machine learning model based on the moving average of the set of parameters calculated during the first iteration.

(96) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for starting a calculation of the moving average of the set of parameters after a configured number of iterations from a starting iteration of the set of multiple iterations.

(97) Some examples of the method, apparatuses, and non-transitory computer-readable medium described herein may further include operations, features, means, or instructions for determining, for a first iteration, a first validation performance value associated with a first state of a machine learning model of the set of multiple machine learning models, determining, for a second iteration following the first iteration, a second validation performance value associated with a second state of the machine learning model of the set of multiple machine learning models, and selecting the first state of the machine learning model for generation of the ensembled machine learning model based on determining that the second validation performance value may be less than the first validation performance value.

(98) In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the set of multiple machine learning models may be trained using a gradient based technique. In some examples of the method, apparatuses, and non-transitory computer-readable medium described herein, the respective outputs from the set of multiple machine learning models includes a respective vector for each machine learning model and the output of the ensembled machine learning model includes the dimension-wise average of the respective vector.

(99) It should be noted that the methods described herein describe possible implementations, and that the operations and the steps may be rearranged or otherwise modified and that other implementations are possible. Furthermore, aspects from two or more of the methods may be combined.

(100) The description set forth herein, in connection with the appended drawings, describes

example configurations and does not represent all the examples that may be implemented or that are within the scope of the claims. The term “exemplary” used herein means “serving as an example, instance, or illustration,” and not “preferred” or “advantageous over other examples.” The detailed description includes specific details for the purpose of providing an understanding of the described techniques. These techniques, however, may be practiced without these specific details. In some instances, well-known structures and devices are shown in block diagram form in order to avoid obscuring the concepts of the described examples.

(101) In the appended figures, similar components or features may have the same reference label. Further, various components of the same type may be distinguished by following the reference label by a dash and a second label that distinguishes among the similar components. If just the first reference label is used in the specification, the description is applicable to any one of the similar components having the same first reference label irrespective of the second reference label.

(102) Information and signals described herein may be represented using any of a variety of different technologies and techniques. For example, data, instructions, commands, information, signals, bits, symbols, and chips that may be referenced throughout the above description may be represented by voltages, currents, electromagnetic waves, magnetic fields or particles, optical fields or particles, or any combination thereof.

(103) The various illustrative blocks and modules described in connection with the disclosure herein may be implemented or performed with a general-purpose processor, a DSP, an ASIC, an FPGA or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general-purpose processor may be a microprocessor, but in the alternative, the processor may be any conventional processor, controller, microcontroller, or state machine. A processor may also be implemented as a combination of computing devices (e.g., a combination of a DSP and a microprocessor, multiple microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration).

(104) The functions described herein may be implemented in hardware, software executed by a processor, firmware, or any combination thereof. If implemented in software executed by a processor, the functions may be stored on or transmitted over as one or more instructions or code on a computer-readable medium. Other examples and implementations are within the scope of the disclosure and appended claims. For example, due to the nature of software, functions described herein can be implemented using software executed by a processor, hardware, firmware, hardwiring, or combinations of any of these. Features implementing functions may also be physically located at various positions, including being distributed such that portions of functions are implemented at different physical locations. Also, as used herein, including in the claims, “or” as used in a list of items (for example, a list of items prefaced by a phrase such as “at least one of” or “one or more of”) indicates an inclusive list such that, for example, a list of at least one of A, B, or C means A or B or C or AB or AC or BC or ABC (i.e., A and B and C). Also, as used herein, the phrase “based on” shall not be construed as a reference to a closed set of conditions. For example, an exemplary step that is described as “based on condition A” may be based on both a condition A and a condition B without departing from the scope of the present disclosure. In other words, as used herein, the phrase “based on” shall be construed in the same manner as the phrase “based at least in part on.”

(105) Computer-readable media includes both non-transitory computer storage media and communication media including any medium that facilitates transfer of a computer program from one place to another. A non-transitory storage medium may be any available medium that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, non-transitory computer-readable media can comprise RAM, ROM, electrically erasable programmable ROM (EEPROM), compact disk (CD) ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other non-transitory medium that can be used

to carry or store desired program code means in the form of instructions or data structures and that can be accessed by a general-purpose or special-purpose computer, or a general-purpose or special-purpose processor. Also, any connection is properly termed a computer-readable medium. For example, if the software is transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. Disk and disc, as used herein, include CD, laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above are also included within the scope of computer-readable media.

(106) The description herein is provided to enable a person skilled in the art to make or use the disclosure. Various modifications to the disclosure will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other variations without departing from the scope of the disclosure. Thus, the disclosure is not limited to the examples and designs described herein, but is to be accorded the broadest scope consistent with the principles and novel features disclosed herein.

Claims

1. A method for machine learning model training, comprising: performing training of a plurality of machine learning models on a first data set associated with a first domain, wherein the plurality of machine learning models comprises respective sets of parameters that are updated across a plurality of iterations during the training, wherein the training comprises, for each machine learning model of the plurality of machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the plurality of iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations; selecting a plurality of model states that are generated during the training of the plurality of machine learning models, wherein the plurality of model states are selected based at least in part on a validation performance of the plurality of model states performed during the training; generating an ensemble machine learning model by aggregating the plurality of machine learning models corresponding to the plurality of selected model states; and performing a machine learning prediction using the ensemble machine learning model on a second data set associated with a second domain different from the first domain, wherein an output of the ensemble machine learning model is a dimension-wise average of respective outputs from the plurality of machine learning models in the ensemble machine learning model.
2. The method of claim 1, further comprising: determining, for one or more iterations of the plurality of iterations, a validation performance value associated with a current state of a machine learning model of the plurality of machine learning models; and selecting a model state for one or more machine learning models of the plurality of machine learning models based on a highest validation performance value for the corresponding machine learning model across all iterations of the plurality of iterations.
3. The method of claim 1, further comprising: determining, for a first iteration following the threshold number of previous iterations, a first set of values for the set of parameters of a machine learning model of the plurality of machine learning models; and determining, for a second iteration following the first iteration, a second set of values for the set of parameters of the machine learning model based at least in part on the moving average of the set of parameters calculated during the first iteration.
4. The method of claim 1, further comprising: starting a calculation of the moving average of the set of parameters after a configured number of iterations from a starting iteration of the plurality of

iterations.

5. The method of claim 1, further comprising: determining, for a first iteration, a first validation performance value associated with a first state of a machine learning model of the plurality of machine learning models; determining, for a second iteration following the first iteration, a second validation performance value associated with a second state of the machine learning model of the plurality of machine learning models; and selecting the first state of the machine learning model for generation of the ensembled machine learning model based at least in part on determining that the second validation performance value is less than the first validation performance value.

6. The method of claim 1, wherein the plurality of machine learning models are trained using a gradient based technique.

7. The method of claim 1, wherein the respective outputs from the plurality of machine learning models comprises a respective vector for each machine learning model and the output of the ensembled machine learning model comprises the dimension-wise average of the respective vector.

8. An apparatus for machine learning model training, comprising: a processor; memory coupled with the processor; and instructions stored in the memory and executable by the processor to cause the apparatus to: perform training of a plurality of machine learning models on a first data set associated with a first domain, wherein the plurality of machine learning models comprises respective sets of parameters that are updated across a plurality of iterations during the training, wherein the training comprises, for each machine learning model of the plurality of machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the plurality of iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations; select a plurality of model states that are generated during the training of the plurality of machine learning models, wherein the plurality of model states are selected based at least in part on a validation performance of the plurality of model states performed during the training; generate an ensembled machine learning model by aggregating the plurality of machine learning models corresponding to the plurality of selected model states; and perform a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain, wherein an output of the ensembled machine learning model is a dimension-wise average of respective outputs from the plurality of machine learning models in the ensembled machine learning model.

9. The apparatus of claim 8, wherein the instructions are further executable by the processor to cause the apparatus to: determine, for one or more iterations of the plurality of iterations, a validation performance value associated with a current state of a machine learning model of the plurality of machine learning models; and select a model state for one or more machine learning models of the plurality of machine learning models based on a highest validation performance value for the corresponding machine learning model across all iterations of the plurality of iterations.

10. The apparatus of claim 8, wherein the instructions are further executable by the processor to cause the apparatus to: determine, for a first iteration following the threshold number of previous iterations, a first set of values for the set of parameters of a machine learning model of the plurality of machine learning models; and determine, for a second iteration following the first iteration, a second set of values for the set of parameters of the machine learning model based at least in part on the moving average of the set of parameters calculated during the first iteration.

11. The apparatus of claim 8, wherein the instructions are further executable by the processor to cause the apparatus to: start a calculation of the moving average of the set of parameters after a configured number of iterations from a starting iteration of the plurality of iterations.

12. The apparatus of claim 8, wherein the instructions are further executable by the processor to cause the apparatus to: determine, for a first iteration, a first validation performance value associated with a first state of a machine learning model of the plurality of machine learning models; determine, for a second iteration following the first iteration, a second validation

performance value associated with a second state of the machine learning model of the plurality of machine learning models; and select the first state of the machine learning model for generation of the ensembled machine learning model based at least in part on determining that the second validation performance value is less than the first validation performance value.

13. The apparatus of claim 8, wherein the plurality of machine learning models are trained using a gradient based technique.

14. The apparatus of claim 8, wherein the respective outputs from the plurality of machine learning models comprises a respective vector for each machine learning model and the output of the ensembled machine learning model comprises the dimension-wise average of the respective vector.

15. A non-transitory computer-readable medium storing code for machine learning model training, the code comprising instructions executable by a processor to: perform training of a plurality of machine learning models on a first data set associated with a first domain, wherein the plurality of machine learning models comprises respective sets of parameters that are updated across a plurality of iterations during the training, wherein the training comprises, for each machine learning model of the plurality of machine learning models, inputting, as values for a set of parameters of the respective sets of parameters and for an iteration of the plurality of iterations, a moving average of the set of parameters calculated over a threshold number of previous iterations; select a plurality of model states that are generated during the training of the plurality of machine learning models, wherein the plurality of model states are selected based at least in part on a validation performance of the plurality of model states performed during the training; generate an ensembled machine learning model by aggregating the plurality of machine learning models corresponding to the plurality of selected model states; and perform a machine learning prediction using the ensembled machine learning model on a second data set associated with a second domain different from the first domain, wherein an output of the ensembled machine learning model is a dimension-wise average of respective outputs from the plurality of machine learning models in the ensembled machine learning model.

16. The non-transitory computer-readable medium of claim 15, wherein the instructions are further executable by the processor to: determine, for one or more iterations of the plurality of iterations, a validation performance value associated with a current state of a machine learning model of the plurality of machine learning models; and select a model state for one or more machine learning models of the plurality of machine learning models based on a highest validation performance value for the corresponding machine learning model across all iterations of the plurality of iterations.

17. The non-transitory computer-readable medium of claim 15, wherein the instructions are further executable by the processor to: determine, for a first iteration following the threshold number of previous iterations, a first set of values for the set of parameters of a machine learning model of the plurality of machine learning models; and determine, for a second iteration following the first iteration, a second set of values for the set of parameters of the machine learning model based at least in part on the moving average of the set of parameters calculated during the first iteration.

18. The non-transitory computer-readable medium of claim 15, wherein the instructions are further executable by the processor to: start a calculation of the moving average of the set of parameters after a configured number of iterations from a starting iteration of the plurality of iterations.

19. The non-transitory computer-readable medium of claim 15, wherein the instructions are further executable by the processor to: determine, for a first iteration, a first validation performance value associated with a first state of a machine learning model of the plurality of machine learning models; determine, for a second iteration following the first iteration, a second validation performance value associated with a second state of the machine learning model of the plurality of machine learning models; and select the first state of the machine learning model for generation of the ensembled machine learning model based at least in part on determining that the second validation performance value is less than the first validation performance value.

20. The non-transitory computer-readable medium of claim 15, wherein the plurality of machine learning models are trained using a gradient based technique.
